Tutorial 3_summary

Tutorial 3 — Dynamic Response, Frequency Analysis & Compensation

> **Source**: Tutorial 3_Solutions.pdf
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1. Step Response of First-Order Systems

1.1 Engineering Background

Used for systems without mass/inertia: thermocouples, liquid-level systems, RC low-pass filter circuits.
These systems only exhibit "resistance" and "capacitance" characteristics.

1.2 Core Parameter: Time Constant τ

Domain	τ Formula
Thermal	$\tau = \dfrac{MC}{UA}$
Electrical	$\tau = RC$

> ✅ Physical meaning: The **smaller** τ, the **faster** the sensor tracks the true signal.

1.3 Mathematical Model — Step Response

When subjected to a step input of height ΔI:

$$\boxed{\Delta O(t) = K \cdot \Delta I \cdot \left(1 - e^{-t/\tau}\right)}$$

Time	Output Reaches
$t = \tau$	**63.2%** of final value
$t = 3\tau$	95.0%
$t = 4\tau \sim 5\tau$	**~99%** → essentially settled

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2. Step Response of Second-Order Systems

2.1 Engineering Background

Used for systems containing inertia: mechanical force sensors, accelerometers.
Due to mass (m), springs (k), and dampers (λ), these systems exhibit **overshoot and oscillation**.

2.2 Two Core Parameters

Parameter	Formula	Physical Meaning
Natural frequency ω_n	$\omega_n = \sqrt{k/m}$	Speed of inherent oscillation
Damping ratio ξ	$\xi = \dfrac{\lambda}{2\sqrt{km}}$	Ability to dissipate energy, suppress oscillation

> ✅ Optimal engineering design: $\xi \approx 0.7$

2.3 Underdamped Step Response ($\xi < 1$)

$$\boxed{\Delta O(t) = K \cdot \Delta I \cdot \left[1 - e^{-\xi\omega_n t}\left(\cos\omega_d t + \frac{\xi}{\sqrt{1-\xi^2}}\sin\omega_d t\right)\right]}$$

where $\omega_d = \omega_n\sqrt{1-\xi^2}$ is the **damped natural frequency**.

2.4 Key Points

Feature	Determined By
Decay envelope	$e^{-\xi\omega_n t}$ → Settling Time $T_s \approx \dfrac{5}{\xi\omega_n}$
Oscillation frequency	Trigonometric terms → ω_d

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3. Handling Complex Periodic Signals — Fourier Series

3.1 The Superposition Principle

> **Fourier's insight**: Any complex **periodic** signal = superposition of sine waves of different frequencies (Fundamental + Higher Harmonics).

For a linear system: **Total Output = Sum of responses to each individual frequency component.**

3.2 General Sinusoidal Response Model

For input $I(t) = A_i \sin(\omega t)$, the steady-state output of a linear system is a sine wave of **the same frequency**:

$$\boxed{O(t) = A_i \cdot |G(j\omega)| \cdot \sin(\omega t + \phi)}$$

Term	Meaning
$\|G(j\omega)\|$	**Amplitude ratio** — determines if signal is amplified or attenuated
ϕ	**Phase shift** — time lag/lead angle

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4. "Frequency Fingerprints" of First and Second-Order Systems

4.1 First-Order Systems (Thermocouples, RC filters)

Characteristic	Formula
Amplitude Ratio	$G(j\omega) = \frac{K}{\sqrt{1+(\omega\tau)^2}}$
Phase Shift	$\phi = -\arctan(\omega\tau)$

> ✅ **Natural Low-pass Filter**: Low frequencies pass with almost no loss ($|G| \approx 1$); high frequencies are severely attenuated.

4.2 Second-Order Systems (Force transducers, Accelerometers)

Characteristic	Formula
Amplitude Ratio	$G(j\omega) = \frac{K}{\sqrt{\left(1-\frac{\omega^2}{\omega_n^2}\right)^2+\left(2\xi\frac{\omega}{\omega_n}\right)^2}}$
Phase Shift	$\phi = -\arctan\left(\frac{2\xi\frac{\omega}{\omega_n}}{1-\frac{\omega^2}{\omega_n^2}}\right)$

> ⚠ **Hidden Resonance Crisis**: When $\omega \approx \omega_n$ and ξ is small, **destructive resonance amplification** occurs → severe measurement distortion.

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5. Open-Loop Dynamic Compensation

5.1 Core Philosophy: Mathematical Cancellation

When physical sensor characteristics cannot be changed (e.g., τ too large → sluggish response), series-connect an electronic **compensation circuit** (lead-lag network) at the output.

5.2 Zero-Pole Cancellation

Component	Transfer Function
Original Sensor	$G_s(s) = \frac{1}{1 + \tau_{slow} s}$
Compensation Circuit	$G_c(s) = \frac{1 + \tau_{slow} s}{1 + \tau_{fast} s}$
Total	$\boxed{G_{total}(s) = G_s(s) \cdot G_c(s) = \frac{1}{1 + \tau_{fast} s}}$

> ✅ **Result**: The slow pole is canceled; replaced by a **much faster** pole $\tau_{fast} \ll \tau_{slow}$.

5.3 Pros & Cons

Pros	Cons
Low cost, simple to implement	**Fatal weakness**: Poor anti-interference
Significantly broadens flat bandwidth	If τ_{slow} drifts (temperature, aging) → cancellation fails completely

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6. Closed-Loop Negative Feedback Architecture

6.1 Core Philosophy: The Ultimate Revolution

> Stop letting the sensor "fight alone." Introduce a **feedback loop** that converts output back to input-domain quantity → compares with original input in real-time → system only responds to the **"net error."**

6.2 Force-Balanced Accelerometer Example

Improvement	Mechanism
Explosive bandwidth & speed	$\omega_{n,c} \gg \omega_{n,open}$ (closed-loop has much higher natural frequency)
Perfect damping	By adjusting circuit gain, ξ_c can be **locked at optimal 0.7**
Ultimate robustness	K no longer depends on aging mechanical parameters (spring stiffness); depends on **highly stable electronic components** (precision resistors) and magnetic field strength

> ✅ **The paradigm shift**: Mechanical precision → Electronic stability. Electrons don't wear out; springs do.

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7. Summary Table — Dynamic Systems

System Type	Key Parameter(s)	Step Response	Frequency Behavior
First-order	τ (time constant)	Exponential approach $1 - e^{-t/\tau}$	Low-pass: $
Second-order, underdamped	$\omega_n, \xi < 1$	Oscillatory decay	Resonance near ω_n
Second-order, critically damped	$\omega_n, \xi = 1$	Fastest non-oscillatory	Smooth roll-off
Second-order, overdamped	$\omega_n, \xi > 1$	Slow, no oscillation	Gradual attenuation
Closed-loop	$\omega_{n,c}, \xi_c$	Tunable via feedback	Extended flat bandwidth

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> ✅ **Key Takeaway**: Open-loop dynamic compensation ($G_c(s)$ cancels $G_s(s)$ pole) is **cheap but fragile**. Closed-loop negative feedback is the **architectural revolution** — it replaces mechanical precision with electronic stability, delivering higher bandwidth, optimal damping, and robustness against aging.